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Algebra Lineal
Grupo 17
Semestre 2022-2

Tarea 7

Del ejercicio visto en clase, obtener M_A^B, sí:

$A = \{(1,2,0), (0,-1,4), (1,-1,1)\}$

$B = \{(2,0,2), (-3,1,1), (0,0,-3)\}$

$$M_B^A = \begin{pmatrix} 7/2 & -3/2 & 1 \\ 2 & -1 & -1 \\ 3 & -8/3 & -4/3 \end{pmatrix}$$

Solución

$$M_A^B = \left([\bar{\omega}_1]_A \; [\bar{\omega}_2]_A \; [\bar{\omega}_3]_A \right)$$

$[\bar{\omega}_1]_A$

$\bar{\omega}_1 = \alpha_1 \bar{v}_1 + \alpha_2 \bar{v}_2 + \alpha_3 \bar{v}_3$

$(2,0,2) = \alpha_1(1,2,0) + \alpha_2(0,-1,4) + \alpha_3(1,-1,1)$

$(2,0,2) = (\alpha_1 + \alpha_3,\ 2\alpha_1 - \alpha_2 - \alpha_3,\ 4\alpha_2 + \alpha_3)$

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$$\alpha_1 + \alpha_3 = 2 \ldots (1)$$
$$2\alpha_1 - \alpha_2 - \alpha_3 = 0 \ldots (2)$$
$$4\alpha_2 + \alpha_3 = 2 \ldots (3)$$

Sistema de ecuaciones

$$\left(\begin{array}{ccc|c} 1 & 0 & 1 & 2 \\ 2 & -1 & -1 & 0 \\ 0 & 4 & 1 & 2 \end{array}\right) R_1(-2) + R_2 = R_2$$

$$\left(\begin{array}{ccc|c} 1 & 0 & 1 & 2 \\ 0 & -1 & -3 & -4 \\ 0 & 4 & 1 & 2 \end{array}\right) R_2(-1)$$

$$\left(\begin{array}{ccc|c} 1 & 0 & 1 & 2 \\ 0 & 1 & 3 & 4 \\ 0 & 4 & 1 & 2 \end{array}\right) R_2(-4) + R_3 = R_3$$

$$\left(\begin{array}{ccc|c} 1 & 0 & 1 & 2 \\ 0 & 1 & 3 & 4 \\ 0 & 0 & -11 & -14 \end{array}\right)$$

$$-11\alpha_3 = -14$$
$$\alpha_3 = \frac{-14}{11}$$

$$\alpha_2 + 3\alpha_3 = 4$$
$$\alpha_2 + 3\left(\frac{-14}{11}\right) = 4$$
$$\alpha_2 = 4 - \frac{42}{11}$$
$$\alpha_2 = \frac{2}{11}$$

$$\alpha_1 + \alpha_3 = 2$$
$$\alpha_1 + \frac{14}{11} = 2$$
$$\alpha_1 = 2 - \frac{14}{11}$$
$$\alpha_1 = \frac{8}{11}$$

Escaneado con CamScanner

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$$[\overline{w}_1]_A = \begin{pmatrix} 8/11 \\ 2/11 \\ 14/11 \end{pmatrix}$$

$[\overline{w}_2]_A$

$\overline{w}_2 = \beta_1 \overline{v}_1 + \beta_2 \overline{v}_2 + \beta_3 \overline{v}_3$

$(-3, 1, 1) = \beta_1 (1, 2, 0) + \beta_2 (0, -1, 4) + \beta_3 (1, -1, 1)$

$(-3, 1, 1) = (\beta_1 + \beta_3,\ 2\beta_1 - \beta_2 - \beta_3,\ 4\beta_2 + \beta_3)$

$$\left.\begin{aligned} \beta_1 + \beta_3 &= -3 \\ 2\beta_1 - \beta_2 - \beta_3 &= 1 \\ 4\beta_2 + \beta_3 &= 1 \end{aligned}\right\} \text{Sistema de ecuaciones}$$

$$\left(\begin{array}{ccc|c} 1 & 0 & 1 & -3 \\ 2 & -1 & -1 & 1 \\ 0 & 4 & 1 & 1 \end{array}\right) R_1(-2) + R_2 = R_2$$

$$\left(\begin{array}{ccc|c} 1 & 0 & 1 & -3 \\ 0 & -1 & -3 & 7 \\ 0 & 4 & 1 & 1 \end{array}\right) R_2(-1)$$

$$\left(\begin{array}{ccc|c} 1 & 0 & 1 & -3 \\ 0 & 1 & 3 & -7 \\ 0 & 4 & 1 & 1 \end{array}\right) R_2(-4) + R_3 = R_3$$

$$\left(\begin{array}{ccc|c} 1 & 0 & 1 & -3 \\ 0 & 1 & 3 & -7 \\ 0 & 0 & -11 & 29 \end{array}\right)$$

$-11\beta_3 = 29$

$\beta_3 = \frac{29}{-11}$ //

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$\beta_2 + 3\beta_3 = -7$

$\beta_2 + 3\left(-\frac{29}{11}\right) = -7$

$\beta_2 = -7 + \frac{87}{11}$

$\beta_2 = \frac{10}{11}$ //

$\beta_1 + \beta_3 = -3$

$\beta_1 + \left(-\frac{29}{11}\right) = -3$

$\beta_1 = -3 + \frac{29}{11}$

$\beta_1 = -\frac{4}{11}$ //

$$[\bar{w}_2]_A = \begin{pmatrix} -4/11 \\ 10/11 \\ -29/11 \end{pmatrix}$$

$[\bar{w}_3]_A$

$\bar{w}_3 = \gamma_1 \bar{v}_1 + \gamma_2 \bar{v}_2 + \gamma_3 \bar{v}_3$

$(0, 0, -3) = \gamma_1(1, 2, 0) + \gamma_2(0, -1, 4) + \gamma_3(1, -1, 1)$

$(0, 0, -3) = (\gamma_1 + \gamma_3, 2\gamma_1 - \gamma_2 - \gamma_3, 4\gamma_2 + \gamma_3)$

$$\left.\begin{aligned} \gamma_1 + \gamma_3 &= 0 \\ 2\gamma_1 - \gamma_2 - \gamma_3 &= 0 \\ 4\gamma_2 + \gamma_3 &= -3 \end{aligned}\right\} \text{Sistema de ecuaciones}$$

$$\left(\begin{array}{ccc|c} 1 & 0 & 1 & 0 \\ 2 & -1 & -1 & 0 \\ 0 & 4 & 1 & -3 \end{array}\right) R_1(-2) + R_2 = R_2$$

$$\left(\begin{array}{ccc|c} 1 & 0 & 1 & 0 \\ 0 & -1 & -3 & 0 \\ 0 & 4 & 1 & -3 \end{array}\right) R_2(-1)$$

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$$\left(\begin{array}{ccc|c} 1 & 0 & 1 & 0 \\ 0 & 1 & 3 & 0 \\ 0 & 4 & 1 & -3 \end{array}\right) R_2(-4)+R_3=R_3$$

$$\left(\begin{array}{ccc|c} 1 & 0 & 1 & 0 \\ 0 & 1 & 3 & 0 \\ 0 & 0 & -11 & -3 \end{array}\right)$$

$-11\gamma_3 = -3$

$\gamma_3 = \frac{3}{11}$ //

$\gamma_2 + 3\gamma_3 = 0$

$\gamma_2 + 3\left(\frac{3}{11}\right) = 0$

$\gamma_2 = -\frac{9}{11}$ //

$\gamma_1 + \gamma_3 = 0$

$\gamma_1 + \frac{3}{11} = 0$

$\gamma_1 = -\frac{3}{11}$ //

$$[\bar{\omega}_3]_A = \begin{pmatrix} -3/11 \\ -9/11 \\ 3/11 \end{pmatrix}$$

$$M_A^B = \begin{pmatrix} \frac{8}{11} & -\frac{4}{11} & -\frac{3}{11} \\ \frac{2}{11} & \frac{10}{11} & -\frac{9}{11} \\ \frac{14}{11} & -\frac{29}{11} & \frac{3}{11} \end{pmatrix}$$